

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad <i>Telangana, India</i> <i>Master of Technology in Computer Science and Engineering</i>	July 2025 – Present <i>CGPA: 9.12</i>
Indian Institute of Information Technology, Ranchi <i>Ranchi, India</i> <i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>	Aug 2020 – Jun 2024 <i>CGPA: 9.05</i>

PROJECTS

Multi-Modal AI-SaaS Platform <i>Next.js 13, TypeScript, Prisma, Stripe, PostgreSQL</i>	🔗
- Engineered a commercial SaaS application orchestrating OpenAI GPT-4 and Replicate APIs for text, code, audio, and image generation. - Implemented customer subscription lifecycles, tiered usage quotas, and secure webhook processing via Stripe SDK and Clerk Auth. - Designed performant database schemas and relational data pipelines using Prisma ORM over cloud-managed PostgreSQL.	
IndicLLM: Transformer Pretraining & Reasoning SFT <i>PyTorch, FlashAttention-2, RoPE</i>	🔗
- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens in Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU. - Trained 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility tax and achieving 0.505 Bits-Per-Byte compression across high and low-resource languages. - Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT.	
BoardBuddy: Real-Time Collaborative Canvas <i>Next.js 14, Convex DB, Liveblocks, TypeScript</i>	🔗
- Architected a real-time collaborative workspace supporting multi-user canvas syncing, freehand sketching, and shape manipulation. - Synchronized real-time state and user presence across concurrent client sessions via Liveblocks WebSockets and Convex reactive DB. - Implemented organization-scoped workspaces and access permissions with Clerk authentication and a responsive TailwindCSS UI.	
Peer-to-Peer Distributed File Sharing System <i>C++, Sockets, Multithreading, OpenSSL</i>	🔗
- Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers. - Engineered out-of-order parallel chunk writing via <code>pwrite()</code> and dynamic bitfield generation for simultaneous partial seeding.	

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Rust, SQL, Bash.

AI & Foundation Models: LLM Orchestration, Prompt Engineering, Transformers (Pretraining & SFT), FlashAttention-2, RoPE, PyTorch.

Full-Stack & Cloud: Next.js 14, React, Node.js, Convex DB, Prisma ORM, PostgreSQL, MongoDB, Stripe SDK, REST APIs.

Systems & Architecture: WebSockets, POSIX Sockets, Multithreading, Concurrency, Linux/Unix, Git, Docker.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

INTERESTS

- Web Novel and Manga Enthusiast · Chess and Badminton.